

## Lab Task#4

*Note: All statements should be commented. Plagiarism will get straight zero.*

### Question 1

A number is called a perfect number if it is equal to the sum of its proper divisors (all positive divisors except the number itself). For example, 6 is perfect because  $1 + 2 + 3 = 6$ .

- Write a program to find and print all perfect numbers between 1 and 500 (inclusive).
- For each candidate number, you must discover its divisors using a loop (do not hardcode divisor lists).
- Use a nested loop structure and a condition to decide when a number is perfect.
- Print each perfect number on a new line.

### Question 2

Compute the value of the following series for n terms:

- $S = x - (x^3)/3! + (x^5)/5! - (x^7)/7! + \dots$  (n terms)
- Use  $x = 1.2$  and  $n = 7$
- Observe the pattern carefully: powers are odd (1, 3, 5, 7, ...), denominators are factorials of those same odd numbers, and signs alternate starting with +
- You must calculate each factorial using a loop (do not use `math.factorial`).
- Print the final value of S.

### Question 3

The Collatz rule for a positive integer m is:

- If m is even:  $m = m / 2$
- If m is odd:  $m = 3 * m + 1$
- Repeat until m becomes 1. The number of updates required is called the step count.
- For every starting value from 1 to 40, compute its Collatz step count.
- Finally print: the starting value that needed the maximum steps, and that maximum step count.
- If two starts share the same maximum steps, print the smaller starting value.